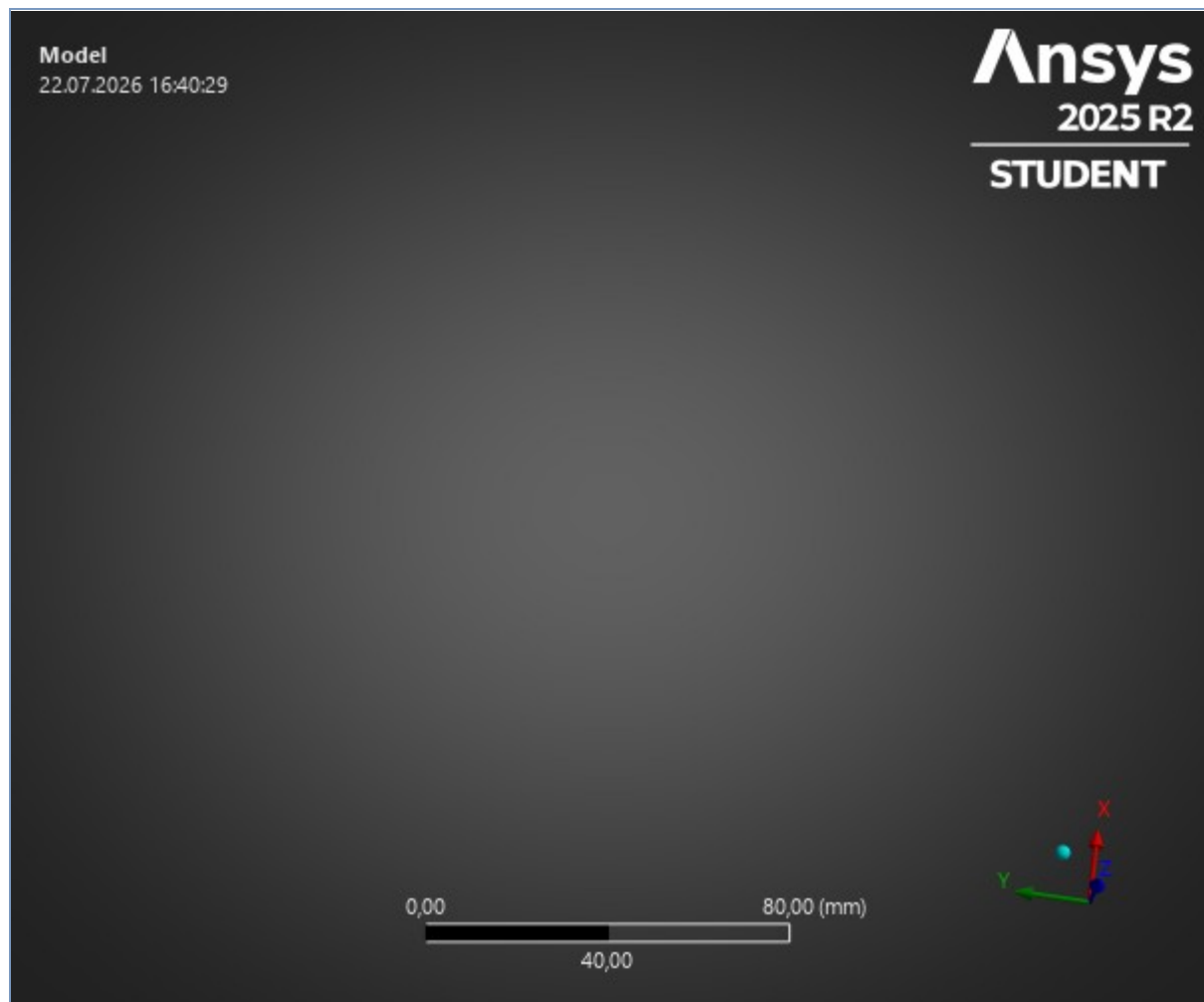




Project

First Saved	Monday, July 20, 2026
Last Saved	Wednesday, July 22, 2026
Product Version	2025 R2
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (B4)

TABLE 2

Model (B4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (B4) > Geometry Imports > Geometry Import (B3)

Object Name	<i>Geometry Import (B3)</i>
State	Solved
Definition	
Source	C:\Users\User\Desktop\Прочностные характеристики\All_work_files\dp0\SYS-1\DM\SYS-1.scdocx

Type	SpaceClaim
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (B4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\User\Desktop\Прочностные характеристики\All_work_files\dp0\SYS-1\DM\SYS-1.scdocx
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	200, mm
Length Y	200, mm
Length Z	50,512 mm
Properties	
Volume	6,9256e+005 mm ³
Mass	0,8257 kg
Scale Factor Value	1,
Statistics	
Bodies	2
Active Bodies	2
Nodes	127233
Elements	70153

Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (B4) > Geometry > Parts

Model B47 - Geometry - Parts		
Object Name	Bтулка\Solid	SYS-1\Твердое тело
State	Meshed	
Graphics Properties		
Visible	Yes	
Transparency	1	
Definition		
Suppressed	No	
Stiffness Behavior	Flexible	
Coordinate System	Default Coordinate System	
Reference Temperature	By Environment	
Treatment	None	
Material		
Assignment	Steel 40X Gost	PA12-CF10
Nonlinear Effects	Yes	
Thermal Strain Effects	Yes	
Bounding Box		
Length X	30,04 mm	200, mm
Length Y	30,04 mm	200, mm
Length Z	49,5 mm	50,512 mm
Properties		
Volume	15483 mm³	6,7708e+005 mm³
Mass	0,12154 kg	0,70416 kg
Centroid X	-1,0201e-015 mm	-3,1788e-004 mm

Centroid Y	-0,61774 mm	-1,5158e-005 mm
Centroid Z	21,75 mm	12,6 mm
Moment of Inertia Ip1	34,891 kg·mm²	1322,2 kg·mm²
Moment of Inertia Ip2	35,835 kg·mm²	1322,2 kg·mm²
Moment of Inertia Ip3	21,119 kg·mm²	2449,5 kg·mm²
Statistics		
Nodes	4113	123120
Elements	731	69422
Mesh Metric	None	
CAD Attributes		
PartTolerance:	0,00000001	
Color:143.149.175		
SCRootPartComponent		

TABLE 6
Model (B4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	4
Material Assignments	0

Coordinate Systems

TABLE 7
Model (B4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0,
Origin	
Origin X	0, mm
Origin Y	0, mm
Origin Z	0, mm
Directional Vectors	
X Axis Data	[1, 0, 0,]
Y Axis Data	[0, 1, 0,]
Z Axis Data	[0, 0, 1,]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (B4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	1
Active Contacts	1

Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (B4) > Connections > Contacts

Object Name	<i>Contacts</i>
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0,
Tolerance Value	0,71829 mm
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75, °
Face Overlap Tolerance	Off
Cylindrical Faces	Include
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	1
Active Connections	1

TABLE 10
Model (B4) > Connections > Contacts > Contact Regions

Object Name	<i>Frictional - SYS-1\Твердое тело To Втулка\Solid</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Contact	2 Faces
Target	1 Face
Contact Bodies	SYS-1\Твердое тело
Target Bodies	Втулка\Solid
Protected	No
Definition	
Type	Frictional
Friction Coefficient	0,2
Scope Mode	Manual
Behavior	Asymmetric
Trim Contact	Off
Contact APDL Name	

Target APDL Name	
Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Stabilization Damping Factor	0,
Pinball Region	Program Controlled
Time Step Controls	None
Geometric Modification	
Interface Treatment	Add Offset, Ramped Effects
Offset	0, mm
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 11
Model (B4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	3,615 mm
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	287,32 mm
Average Surface Area	833,96 mm ²
Minimum Edge Length	2,3673e-004 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5,e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0,272
Maximum Layers	5
Growth Rate	1,2

Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	127233
Elements	70153
Show Detailed Statistics	No

TABLE 12
Model (B4) > Mesh > Mesh Controls

Object Name	<i>Face Sizing</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	60 Faces
Definition	
Suppressed	No
Type	Element Size
Element Size	2,52 mm
Advanced	
Defeature Size	Default
Influence Volume	No
Behavior	Soft

Static Structural (B5)

TABLE 13
Model (B4) > Analysis

Object Name	<i>Static Structural (B5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22, °C
Generate Input Only	No

TABLE 14
Model (B4) > Static Structural (B5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1,

Current Step Number	1,
Step End Time	1, s
Auto Time Stepping	On
Define By	Substeps
Initial Substeps	20,
Minimum Substeps	10,
Maximum Substeps	200,
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Program Controlled
Solver Pivot Checking	Program Controlled
Large Deflection	On
Inertia Relief	Off
Quasi-Static Solution	On
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\User\Desktop\Прочностные характеристики\All_work_files\dp0\SYS-1\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes

Solver Units	Active System
Solver Unit System	mm

TABLE 15
Model (B4) > Static Structural (B5) > Rotations

Object Name	<i>Rotational Velocity</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Define By	Components
Coordinate System	Global Coordinate System
X Component	0, rad/s (ramped)
Y Component	0, rad/s (ramped)
Z Component	-1151,9 rad/s (ramped)
X Coordinate	1,0191e-015 mm
Y Coordinate	-0,61961 mm
Z Coordinate	46,5 mm
Suppressed	No

FIGURE 1
Model (B4) > Static Structural (B5) > Rotational Velocity

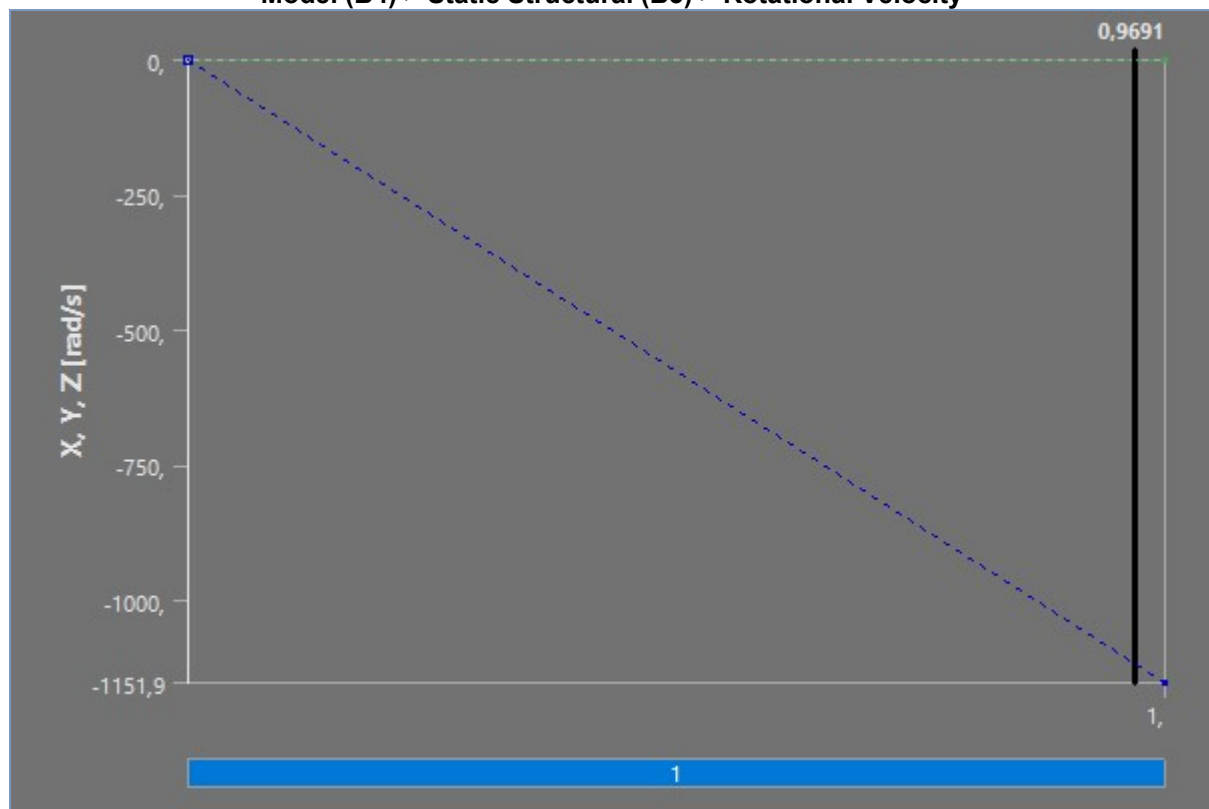


TABLE 16
Model (B4) > Static Structural (B5) > Loads

Object Name	<i>Thermal Condition</i>	<i>Pressure</i>	<i>Cylindrical Support</i>	<i>Remote Displacement</i>
State	Fully Defined		Suppressed	Fully Defined
Scope				
Scoping Method	Geometry Selection			
Geometry	2 Bodies	60 Faces	1 Face	
Coordinate System				Global Coordinate System

X Coordinate				2,7818e-016 mm
Y Coordinate				2,6427e-016 mm
Z Coordinate				21,75 mm
Location				Defined
Definition				
Type	Thermal Condition	Pressure	Cylindrical Support	Remote Displacement
Magnitude	22, °C (ramped)	0,10132 MPa (ramped)		
Suppressed		No	Yes	No
Define By		Normal To		
Applied By		Surface Effect		
Loaded Area		Deformed		
Radial			Fixed	
Axial			Fixed	
Tangential			Fixed	
X Component				0, mm (ramped)
Y Component				0, mm (ramped)
Z Component				0, mm (ramped)
Rotation X				0, ° (ramped)
Rotation Y				0, ° (ramped)
Rotation Z				0, ° (ramped)
Behavior				Deformable
Advanced				
Pinball Region				All

FIGURE 2
Model (B4) > Static Structural (B5) > Thermal Condition

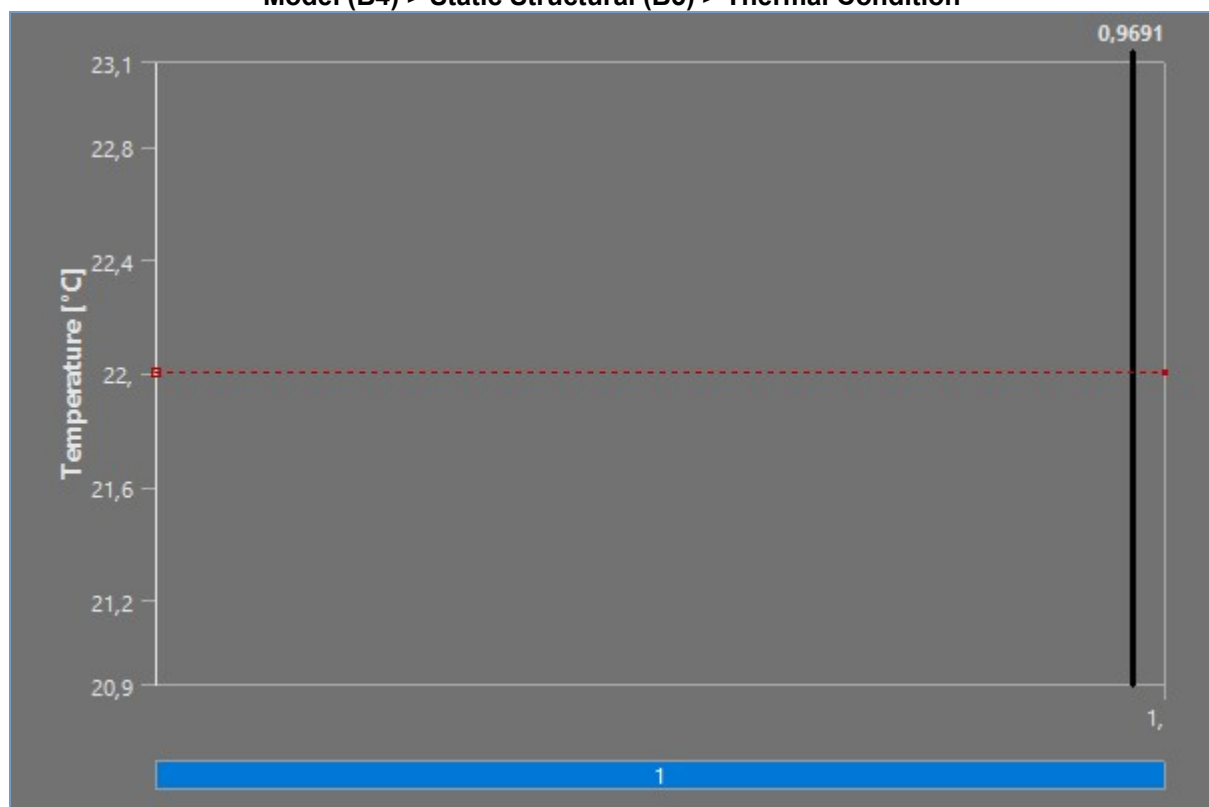


FIGURE 3
Model (B4) > Static Structural (B5) > Pressure

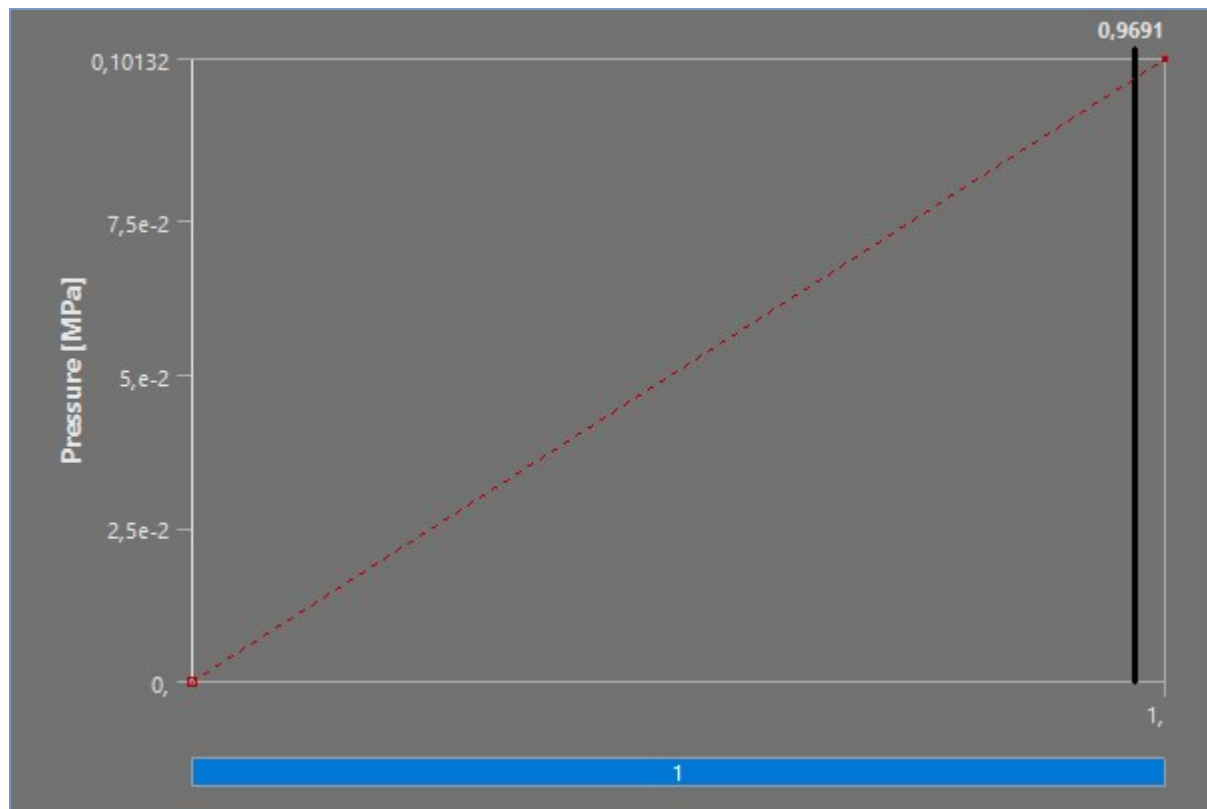
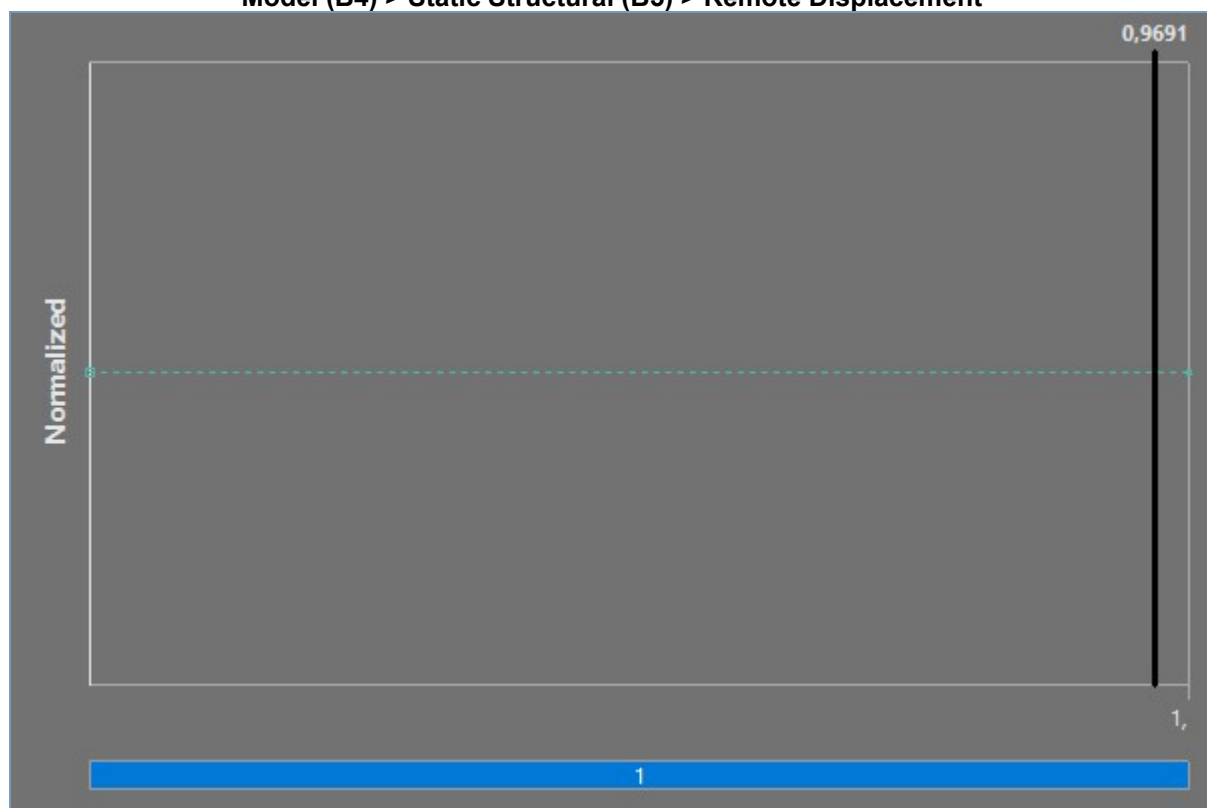


FIGURE 4
Model (B4) > Static Structural (B5) > Remote Displacement



Solution (B6)

TABLE 17
Model (B4) > Static Structural (B5) > Solution

Object Name	<i>Solution (B6)</i>
State	Solved

Adaptive Mesh Refinement	
Max Refinement Loops	1,
Refinement Depth	2,
Information	
Status	Done
MAPDL Elapsed Time	3 h 42 m
MAPDL Memory Used	5,6309 GB
MAPDL Result File Size	3,0167 GB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 18
Model (B4) > Static Structural (B5) > Solution (B6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	5
Update Interval	2,5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 19
Model (B4) > Static Structural (B5) > Solution (B6) > Results

Object Name	Total Deformation	Equivalent Stress	Equivalent Stress 2
State	Solved		
Scope			
Scoping Method	Geometry Selection		
Geometry	All Bodies	148 Faces	
Definition			
Type	Total Deformation	Equivalent (von-Mises) Stress	
By	Time		
Display Time	Last		
Separate Data by Entity	No		
Calculate Time History	Yes		
Identifier			
Suppressed	No		
Results			
Minimum	1,4279e-004 mm	1,7846e-002 MPa	
Maximum	1,0777e+005 mm	10,986 MPa	
Average	1,0429e+005 mm	2,1044 MPa	2,3405 MPa
Minimum Occurs On	Втулка\Solid	SYS-1\Твердое тело	
Maximum Occurs On	SYS-1\Твердое тело		
Minimum Value Over Time			
Minimum	1,2398e-006 mm	7,3146e-005 MPa	
Maximum	1,4279e-004 mm	1,7846e-002 MPa	
Maximum Value Over Time			

Minimum	1,4328e-003 mm	1,9866 MPa	0,23765 MPa
Maximum	1,0777e+005 mm	15,439 MPa	10,986 MPa
Information			
Time	1, s		
Load Step	1		
Substep	84		
Iteration Number	1024		
Integration Point Results			
Display Option		Averaged	
Average Across Bodies		No	

FIGURE 5
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation

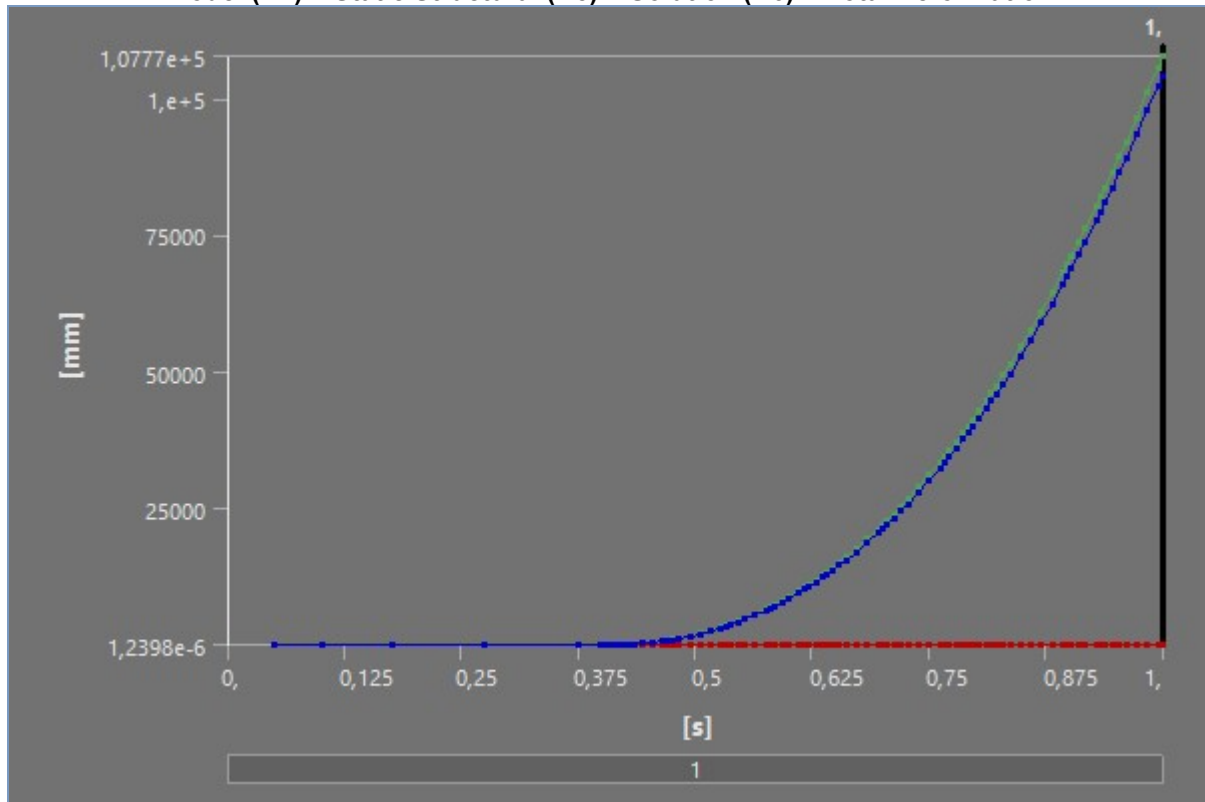


TABLE 20
Model (B4) > Static Structural (B5) > Solution (B6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
5,e-002	1,2398e-006	1,4328e-003	1,0855e-003
0,1	2,2123e-006	2,9874e-003	2,0608e-003
0,175	1,7719e-006	6,2115e-003	3,4405e-003
0,275	3,0775e-006	1,3459e-002	5,4998e-003
0,375	1,2482e-005	2,4413e-002	8,2398e-003
0,3975	1,425e-005	2,6847e-002	9,1412e-003
0,4025	2,0849e-006	1,0426	0,97608
0,4075	2,2201e-006	4,2316	4,004
0,4125	1,1519e-005	10,612	10,103
0,4175	1,1515e-005	21,418	20,468
0,4225	1,004e-005	38,019	36,283
0,4275	2,6089e-005	65,218	62,57
0,435	2,7002e-005	130,82	126,
0,4425	2,7931e-005	221,78	213,87
0,45375	2,9355e-005	416,49	402,03
0,465	3,0814e-005	670,76	647,87

0,47	3,1482e-005	795,7	768,66
0,475	3,2157e-005	932,68	901,11
0,4825	3,3183e-005	1165,6	1126,4
0,49375	3,4753e-005	1578,4	1525,5
0,505	3,6358e-005	2055,8	1987,3
0,51625	3,7999e-005	2599,2	2513,
0,5275	3,9677e-005	3210,	3104,1
0,5325	4,0434e-005	3494,9	3379,8
0,5375	4,1199e-005	3793,4	3668,6
0,545	4,2359e-005	4272,1	4131,9
0,5525	4,3535e-005	4782,2	4625,6
0,56375	4,5329e-005	5619,4	5435,9
0,575	4,7159e-005	6530,1	6317,4
0,58	4,7984e-005	6949,5	6723,4
0,585	4,8816e-005	7383,7	7143,6
0,5925	5,0078e-005	8068,5	7806,5
0,6	5,1356e-005	8787,5	8502,5
0,61125	5,3303e-005	9944,1	9622,
0,61625	5,4179e-005	10474	10135
0,62125	5,5063e-005	11019	10662
0,62875	5,6403e-005	11873	11489
0,63625	5,7758e-005	12763	12350
0,64125	5,8671e-005	13372	12940
0,64625	5,9591e-005	13998	13545
0,65375	6,0984e-005	14974	14490
0,66125	6,2393e-005	15987	15470
0,6725	6,4536e-005	17593	17024
0,68375	6,6716e-005	19286	18663
0,695	6,8932e-005	21068	20387
0,7	6,9929e-005	21877	21170
0,705	7,0932e-005	22705	21970
0,7125	7,2451e-005	23986	23210
0,72	7,3986e-005	25307	24489
0,7275	7,5537e-005	26670	25808
0,73875	7,7893e-005	28809	27877
0,75	8,0286e-005	31042	30039
0,76125	8,2715e-005	33373	32293
0,76625	8,3806e-005	34427	33314
0,77125	8,4904e-005	35502	34354
0,77875	8,6565e-005	37157	35955
0,78625	8,8242e-005	38856	37600
0,79125	8,9369e-005	40009	38716
0,79625	9,0503e-005	41182	39850
0,80375	9,2217e-005	42986	41597
0,81125	9,3948e-005	44836	43387
0,81625	9,511e-005	46090	44600
0,82125	9,628e-005	47365	45833
0,82875	9,8048e-005	49323	47728
0,83625	9,9832e-005	51328	49669
0,8475	1,0254e-004	54443	52683
0,85875	1,0528e-004	57666	55802
0,87	1,0806e-004	60999	59028
0,88125	1,1087e-004	64444	62361
0,8925	1,1372e-004	68001	65802
0,8975	1,15e-004	69604	67354
0,9025	1,1629e-004	71230	68927
0,91			

	1,1823e-004	73719	71336
0,9175	1,2019e-004	76260	73794
0,92875	1,2315e-004	80187	77595
0,93375	1,2448e-004	81956	79307
0,93875	1,2582e-004	83748	81041
0,94625	1,2784e-004	86489	83693
0,95375	1,2988e-004	89284	86397
0,96125	1,3193e-004	92131	89153
0,9725	1,3504e-004	96525	93405
0,98375	1,3818e-004	1,0104e+005	97776
0,995	1,4136e-004	1,0568e+005	1,0227e+005
1,	1,4279e-004	1,0777e+005	1,0429e+005

FIGURE 6
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

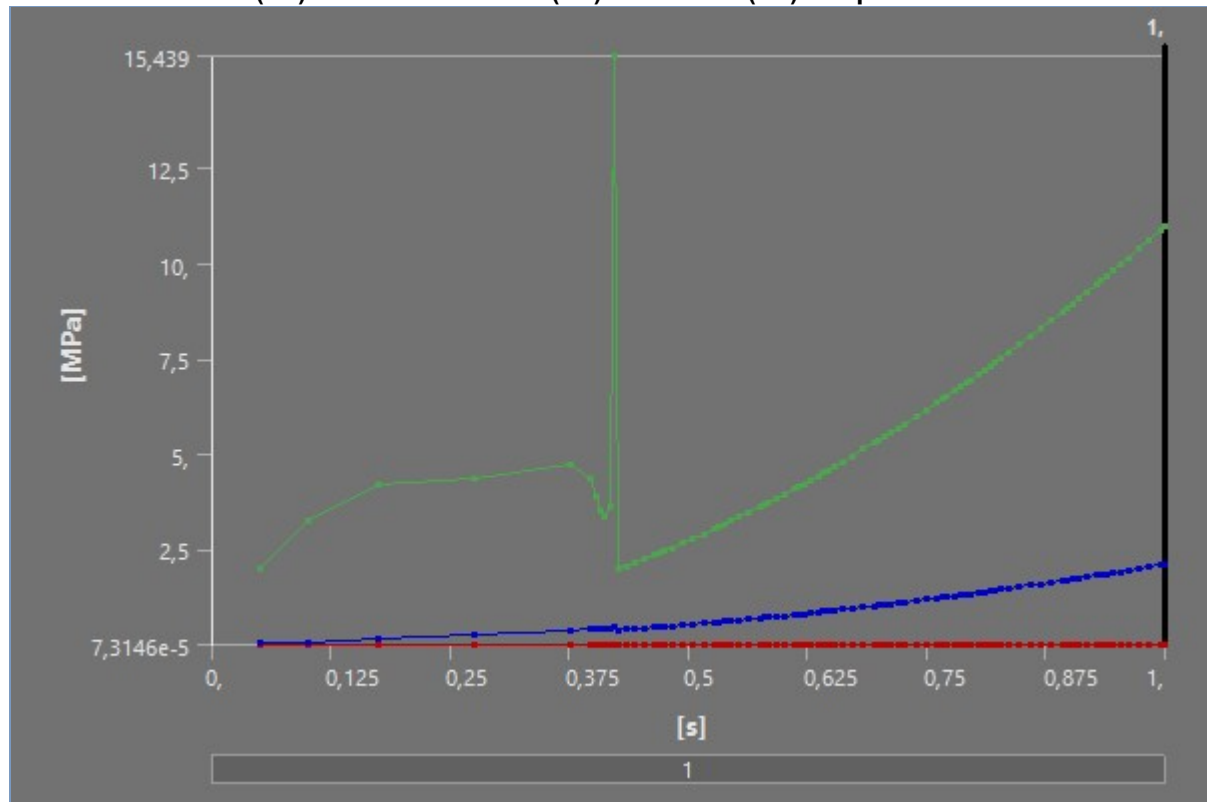


TABLE 21
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
5,e-002	7,3146e-005	1,9964	3,5494e-002
0,1	2,2397e-004	3,2765	7,2371e-002
0,175	3,6712e-004	4,2101	0,13727
0,275	8,8443e-004	4,3543	0,24391
0,375	1,8378e-003	4,7236	0,37386
0,3975	2,004e-003	4,3431	0,40014
0,4025	2,0147e-003	3,8929	0,40027
0,4075	2,1143e-003	3,5418	0,40188
0,4125	2,2809e-003	3,3372	0,40633
0,4175	2,3603e-003	3,6235	0,41341
0,4225	1,6322e-003	15,439	0,45151
0,4275	2,5876e-003	1,9866	0,38542
0,435	2,703e-003	2,0574	0,39897
0,4425	2,8822e-003	2,1298	0,41287

0,45375	2,9273e-003	2,2404	0,43415
0,465	3,2618e-003	2,3534	0,45632
0,47	3,144e-003	2,4069	0,466
0,475	3,2452e-003	2,4578	0,47569
0,4825	3,5754e-003	2,5366	0,49149
0,49375	3,495e-003	2,6572	0,51402
0,505	3,7197e-003	2,7818	0,53799
0,51625	3,9879e-003	2,9085	0,56232
0,5275	4,4284e-003	3,0384	0,58854
0,5325	4,4132e-003	3,0982	0,59852
0,5375	4,5706e-003	3,154	0,60894
0,545	4,6835e-003	3,2429	0,62594
0,5525	4,9027e-003	3,3348	0,64361
0,56375	5,3755e-003	3,4724	0,67036
0,575	5,6832e-003	3,6146	0,69757
0,58	5,6618e-003	3,6829	0,71047
0,585	5,5391e-003	3,7418	0,72162
0,5925	5,8065e-003	3,838	0,73994
0,6	6,3535e-003	3,9388	0,75973
0,61125	6,4307e-003	4,0867	0,78789
0,61625	6,3227e-003	4,1615	0,80168
0,62125	6,4632e-003	4,2237	0,8137
0,62875	6,893e-003	4,3259	0,83367
0,63625	7,0567e-003	4,4321	0,85397
0,64125	6,8828e-003	4,5046	0,86726
0,64625	6,9775e-003	4,573	0,88028
0,65375	7,2085e-003	4,6786	0,90065
0,66125	7,3746e-003	4,7891	0,92172
0,6725	7,8068e-003	4,952	0,95326
0,68375	8,2441e-003	5,1215	0,98602
0,695	8,2489e-003	5,2904	1,019
0,7	8,2747e-003	5,3711	1,0327
0,705	8,4189e-003	5,443	1,0473
0,7125	8,9486e-003	5,5596	1,0698
0,72	8,8011e-003	5,6789	1,0916
0,7275	9,2115e-003	5,7998	1,1156
0,73875	9,4895e-003	5,9787	1,1493
0,75	9,7908e-003	6,1651	1,1855
0,76125	9,6867e-003	6,3483	1,2205
0,76625	1,0067e-002	6,4382	1,2364
0,77125	1,0156e-002	6,5175	1,2524
0,77875	1,0587e-002	6,6449	1,2765
0,78625	1,0659e-002	6,776	1,3011
0,79125	1,0878e-002	6,8664	1,3185
0,79625	1,1036e-002	6,9515	1,3348
0,80375	1,1442e-002	7,0826	1,3605
0,81125	1,1338e-002	7,2191	1,3882
0,81625	1,1533e-002	7,3082	1,4026
0,82125	1,1809e-002	7,3964	1,4197
0,82875	1,22e-002	7,5313	1,446
0,83625	1,2089e-002	7,6724	1,4748
0,8475	1,269e-002	7,8761	1,5121
0,85875	1,3114e-002	8,0889	1,5524
0,87	1,2954e-002	8,3035	1,5944
0,88125	1,3458e-002	8,5197	1,6353
0,8925	1,3554e-002	8,7271	1,6745
0,90375			

	1,4286e-002	8,8418	1,6953
0,9025	1,4356e-002	8,9351	1,7145
0,91	1,4663e-002	9,0836	1,7419
0,9175	1,4646e-002	9,2384	1,7723
0,92875	1,4937e-002	9,463	1,8161
0,93375	1,5381e-002	9,5729	1,8345
0,93875	1,5662e-002	9,6706	1,8537
0,94625	1,5841e-002	9,826	1,8843
0,95375	1,6046e-002	9,9845	1,9133
0,96125	1,6284e-002	10,146	1,9472
0,9725	1,6806e-002	10,38	1,99
0,98375	1,7114e-002	10,625	2,037
0,995	1,7241e-002	10,87	2,0839
1,	1,7846e-002	10,986	2,1044

FIGURE 7
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress 2

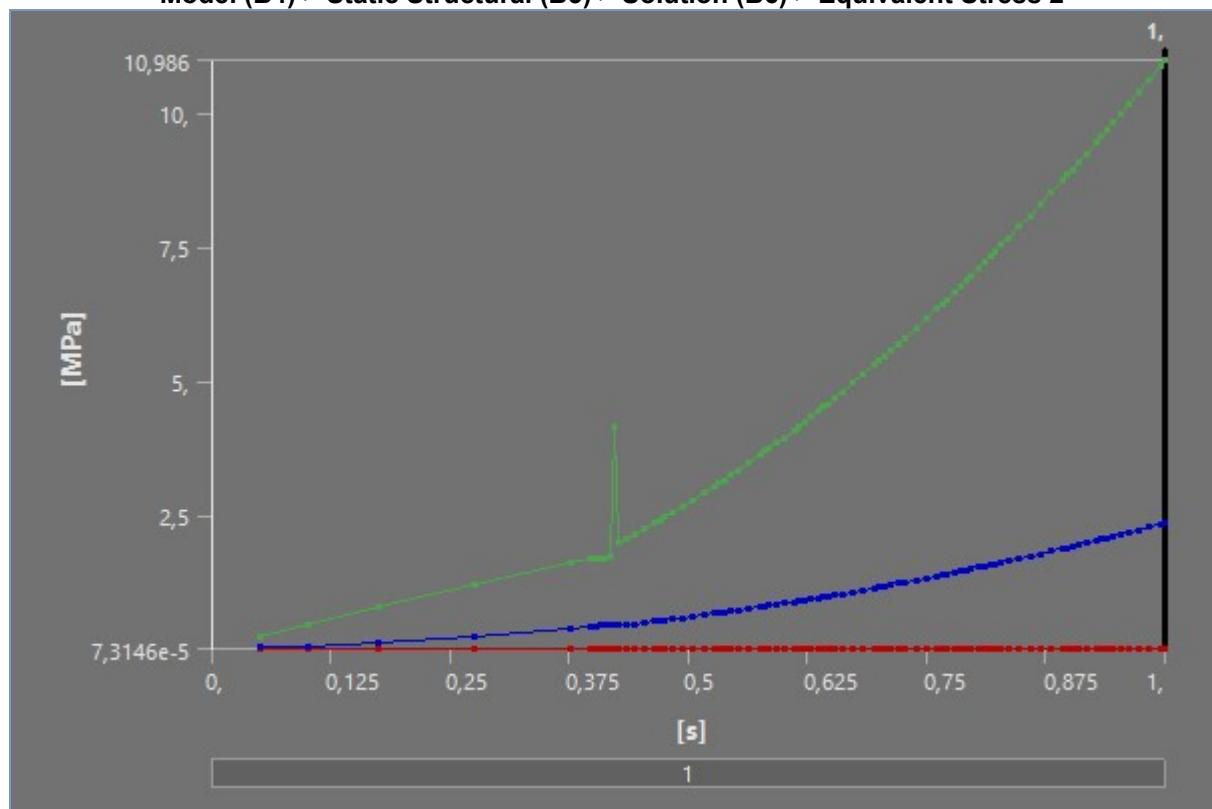


TABLE 22
Model (B4) > Static Structural (B5) > Solution (B6) > Equivalent Stress 2

Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
5,e-002	7,3146e-005	0,23765	2,0517e-002
0,1	2,2397e-004	0,45237	4,8273e-002
0,175	3,6712e-004	0,77114	0,10802
0,275	8,8443e-004	1,197	0,22325
0,375	1,8378e-003	1,6209	0,37955
0,3975	2,004e-003	1,6845	0,41792
0,4025	2,0147e-003	1,6709	0,42403
0,4075	2,1143e-003	1,6684	0,43017
0,4125	2,2809e-003	1,6835	0,43659
0,4175	2,3603e-003	1,7268	0,44262
0,4225	1,6322e-003	4,1404	0,46066
0,4275	2,5876e-003	1,9866	0,42996

0,435	2,703e-003	2,0574	0,44504
0,4425	2,8822e-003	2,1298	0,4605
0,45375	2,9273e-003	2,2404	0,48414
0,465	3,2618e-003	2,3534	0,50976
0,47	3,144e-003	2,4069	0,51958
0,475	3,2452e-003	2,4578	0,53037
0,4825	3,5754e-003	2,5366	0,54826
0,49375	3,495e-003	2,6572	0,57295
0,505	3,7197e-003	2,7818	0,59957
0,51625	3,9879e-003	2,9085	0,62661
0,5275	4,4284e-003	3,0384	0,65705
0,5325	4,4132e-003	3,0982	0,66692
0,5375	4,5706e-003	3,154	0,67861
0,545	4,6835e-003	3,2429	0,69741
0,5525	4,9027e-003	3,3348	0,71705
0,56375	5,3755e-003	3,4724	0,74665
0,575	5,6832e-003	3,6146	0,77689
0,58	5,6618e-003	3,6829	0,79137
0,585	5,5391e-003	3,7418	0,804
0,5925	5,8065e-003	3,838	0,82411
0,6	6,3535e-003	3,9388	0,84595
0,61125	6,4307e-003	4,0867	0,87732
0,61625	6,3227e-003	4,1615	0,89283
0,62125	6,4632e-003	4,2237	0,90611
0,62875	6,893e-003	4,3259	0,92818
0,63625	7,0567e-003	4,4321	0,95078
0,64125	6,8828e-003	4,5046	0,96571
0,64625	6,9775e-003	4,573	0,98017
0,65375	7,2085e-003	4,6786	1,0028
0,66125	7,3746e-003	4,7891	1,0262
0,6725	7,8068e-003	4,952	1,0612
0,68375	8,2441e-003	5,1215	1,0974
0,695	8,2489e-003	5,2904	1,1349
0,7	8,2747e-003	5,3711	1,1497
0,705	8,4189e-003	5,443	1,1662
0,7125	8,9486e-003	5,5596	1,1906
0,72	8,8011e-003	5,6789	1,2151
0,7275	9,2115e-003	5,7998	1,2415
0,73875	9,4895e-003	5,9787	1,2791
0,75	9,7908e-003	6,1651	1,3192
0,76125	9,6867e-003	6,3483	1,3598
0,76625	1,0067e-002	6,4382	1,376
0,77125	1,0156e-002	6,5175	1,3942
0,77875	1,0587e-002	6,6449	1,4205
0,78625	1,0659e-002	6,776	1,4479
0,79125	1,0878e-002	6,8664	1,4673
0,79625	1,1036e-002	6,9515	1,4853
0,80375	1,1442e-002	7,0826	1,5137
0,81125	1,1338e-002	7,2191	1,5464
0,81625	1,1533e-002	7,3082	1,5608
0,82125	1,1809e-002	7,3964	1,5797
0,82875	1,22e-002	7,5313	1,6088
0,83625	1,2089e-002	7,6724	1,6427
0,8475	1,269e-002	7,8761	1,6822
0,85875	1,3114e-002	8,0889	1,7271
0,87	1,2954e-002	8,3035	1,7734
0,88125			

	1,3458e-002	8,5197	1,819
0,8925	1,3554e-002	8,7271	1,865
0,8975	1,4286e-002	8,8418	1,8859
0,9025	1,4356e-002	8,9351	1,9079
0,91	1,4663e-002	9,0836	1,9376
0,9175	1,4646e-002	9,2384	1,9711
0,92875	1,4937e-002	9,463	2,0218
0,93375	1,5381e-002	9,5729	2,0407
0,93875	1,5662e-002	9,6706	2,0619
0,94625	1,5841e-002	9,826	2,0957
0,95375	1,6046e-002	9,9845	2,1282
0,96125	1,6284e-002	10,146	2,168
0,9725	1,6806e-002	10,38	2,2132
0,98375	1,7114e-002	10,625	2,2655
0,995	1,7241e-002	10,87	2,3173
1,	1,7846e-002	10,986	2,3405

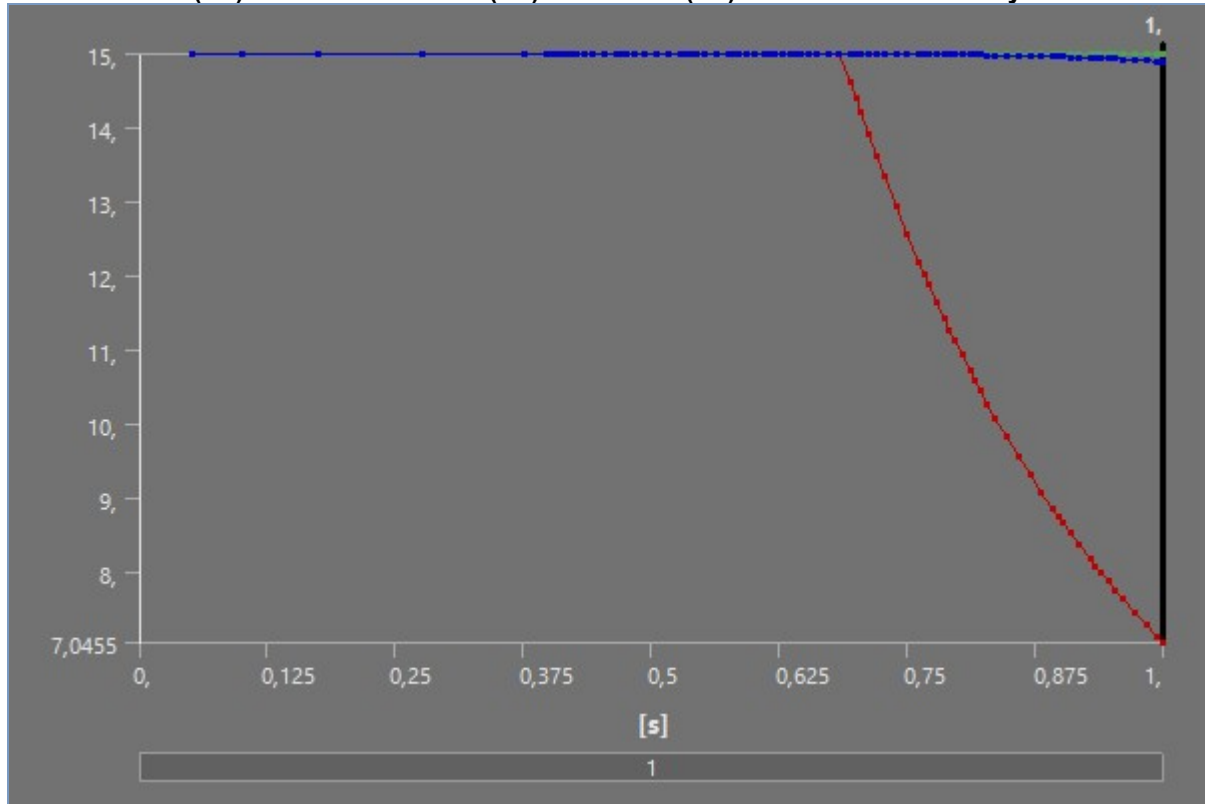
TABLE 23**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Safety Tools**

Object Name	<i>Stress Tool</i>
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 24**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Results**

Object Name	<i>Safety Factor</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	7,0455
Minimum Occurs On	SYS-1\Твердое тело
Minimum Value Over Time	
Minimum	7,0455
Maximum	15,
Maximum Value Over Time	
Minimum	15,
Maximum	15,
Information	
Time	1, s
Load Step	1
Substep	84

Iteration Number	1024
------------------	------

FIGURE 8**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Safety Factor****TABLE 25****Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool > Safety Factor**

Time [s]	Minimum	Maximum	Average
5,e-002	15,		15,
0,1			
0,175			
0,275			
0,375			
0,3975			
0,4025			
0,4075			
0,4125			
0,4175			
0,4225			
0,4275			
0,435			
0,4425			
0,45375			
0,465			
0,47			
0,475			
0,4825			
0,49375			
0,505			
0,51625			
0,5275			
0,5325			
0,5375			

0,545		
0,5525		
0,56375		
0,575		
0,58		
0,585		
0,5925		
0,6		
0,61125		
0,61625		
0,62125		
0,62875		
0,63625		
0,64125		
0,64625		
0,65375		
0,66125		
0,6725		
0,68375		
0,695	14,63	
0,7	14,41	
0,705	14,22	
0,7125	13,922	
0,72	13,629	14,999
0,7275	13,345	
0,73875	12,946	14,998
0,75	12,554	14,997
0,76125	12,192	14,996
0,76625	12,022	14,995
0,77125	11,876	
0,77875	11,648	14,994
0,78625	11,423	14,993
0,79125	11,272	14,992
0,79625	11,134	14,991
0,80375	10,928	14,99
0,81125	10,722	14,988
0,81625	10,591	14,987
0,82125	10,465	14,985
0,82875	10,277	14,983
0,83625	10,088	14,981
0,8475	9,8272	14,978
0,85875	9,5687	14,974
0,87	9,3213	14,971
0,88125	9,0848	14,967
0,8925	8,8689	14,964
0,8975	8,7539	14,962
0,9025	8,6625	14,959
0,91	8,5208	14,955
0,9175	8,3781	14,949
0,92875	8,1792	14,946
0,93375	8,0853	14,944
0,93875	8,0036	14,939
0,94625	7,8771	14,934
0,95375	7,752	14,927
0,96125	7,6288	14,919
0,9725	7,4568	

15,

0,98375	7,285	14,909
0,995	7,1208	14,898
1,	7,0455	14,893

TABLE 26**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Safety Tools**

Object Name	<i>Stress Tool 2</i>
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 27**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool 2 > Results**

Object Name	<i>Safety Factor</i>
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	148 Faces
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Separate Data by Entity	No
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	7,0455
Minimum Occurs On	SYS-1\Твердое тело
Minimum Value Over Time	
Minimum	7,0455
Maximum	15,
Maximum Value Over Time	
Minimum	15,
Maximum	15,
Information	
Time	1, s
Load Step	1
Substep	84
Iteration Number	1024

FIGURE 9**Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool 2 > Safety Factor**

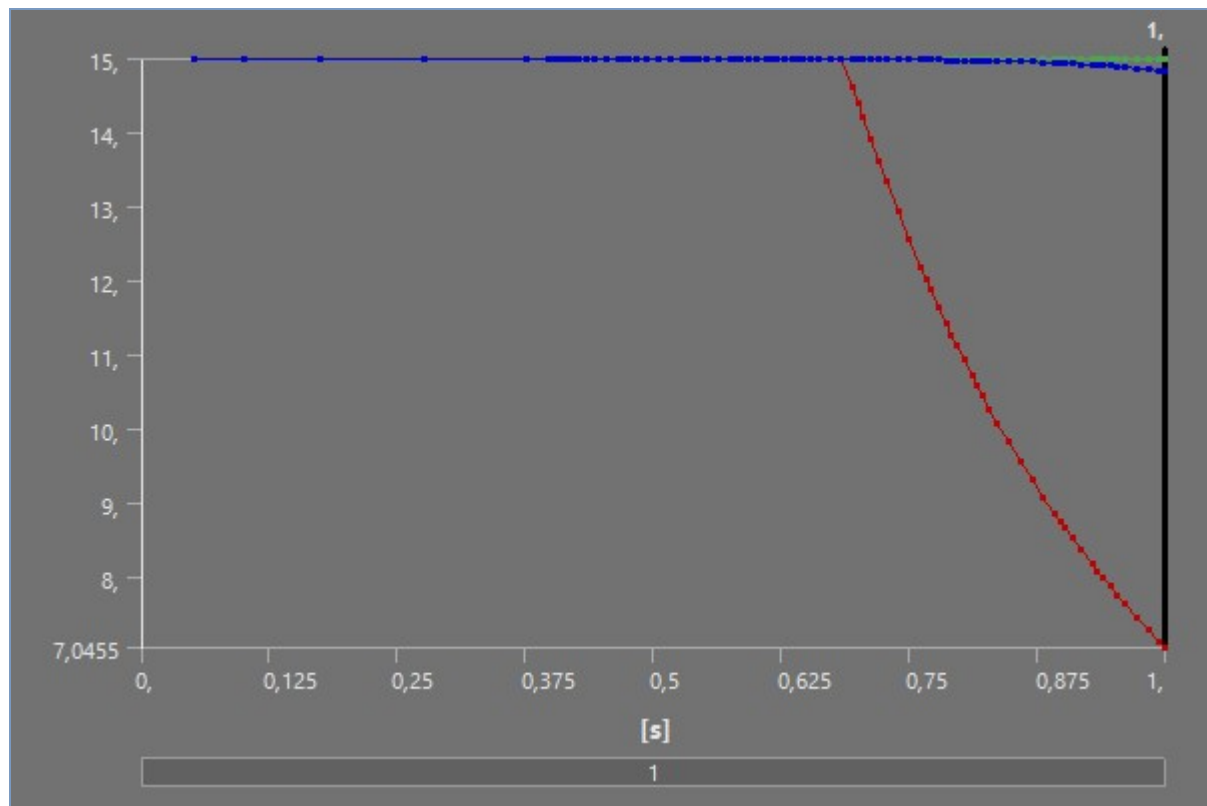


TABLE 28
Model (B4) > Static Structural (B5) > Solution (B6) > Stress Tool 2 > Safety Factor

Time [s]	Minimum	Maximum	Average
5,e-002			
0,1			
0,175			
0,275			
0,375			
0,3975			
0,4025			
0,4075			
0,4125			
0,4175			
0,4225			
0,4275			
0,435			
0,4425			
0,45375	15,		15,
0,465			
0,47			
0,475			
0,4825			
0,49375			
0,505			
0,51625			
0,5275			
0,5325			
0,5375			
0,545			
0,5525			
0,56375			
0,575			

0,58			
0,585			
0,5925			
0,6			
0,61125			
0,61625			
0,62125			
0,62875			
0,63625			
0,64125			
0,64625			
0,65375			
0,66125			
0,6725			
0,68375			
0,695	14,63		
0,7	14,41		
0,705	14,22		
0,7125	13,922		14,999
0,72	13,629		14,998
0,7275	13,345		14,997
0,73875	12,946		14,996
0,75	12,554		14,994
0,76125	12,192		14,992
0,76625	12,022		14,991
0,77125	11,876		14,99
0,77875	11,648		14,988
0,78625	11,423	15,	14,986
0,79125	11,272		14,985
0,79625	11,134		14,984
0,80375	10,928		14,982
0,81125	10,722		14,98
0,81625	10,591		14,978
0,82125	10,465		14,977
0,82875	10,277		14,975
0,83625	10,088		14,972
0,8475	9,8272		14,968
0,85875	9,5687		14,964
0,87	9,3213		14,96
0,88125	9,0848		14,954
0,8925	8,8689		14,949
0,8975	8,7539		14,945
0,9025	8,6625		14,942
0,91	8,5208		14,938
0,9175	8,3781		14,932
0,92875	8,1792		14,922
0,93375	8,0853		14,919
0,93875	8,0036		14,914
0,94625	7,8771		14,906
0,95375	7,752		14,898
0,96125	7,6288		14,888
0,9725	7,4568		14,874
0,98375	7,285		14,858
0,995	7,1208		14,839
1,	7,0455		14,83

Material Data

Steel 40X Gost

TABLE 29
Steel 40X Gost > Constants

Density	7,85e-006 kg mm ⁻³
Coefficient of Thermal Expansion	1,2e-005 K ⁻¹

TABLE 30
Steel 40X Gost > Color

Red	Green	Blue
181,	155,	130,

TABLE 31
Steel 40X Gost > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
2,e+005	0,3	1,6667e+005	76923	

TABLE 32
Steel 40X Gost > Tensile Yield Strength

Tensile Yield Strength MPa
785,

TABLE 33
Steel 40X Gost > Compressive Yield Strength

Compressive Yield Strength MPa
785,

TABLE 34
Steel 40X Gost > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
980,

PA12-CF10

TABLE 35
PA12-CF10 > Constants

Density	1,04e-006 kg mm ⁻³
Coefficient of Thermal Expansion	7,e-005 K ⁻¹

TABLE 36
PA12-CF10 > Color

Red	Green	Blue
235,	222,	222,

TABLE 37
PA12-CF10 > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
3311,	0,38	4598,6	1199,6	

TABLE 38
PA12-CF10 > Tensile Yield Strength

Tensile Yield Strength MPa
77,4

TABLE 39

PA12-CF10 > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
90,